Daniel Peelen

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Education

University of Pittsburgh September 2023 - May 2027

B.S. Computer Science, Hungarian Minor — 3.9 GPA, Dean's List

- Coursework: Deep Learning, Data-Driven Systems, Data Structures & Algorithms, Systems Software

Experience

Microsoft May 2026 – August 2026

Software Engineering Intern *Redmond, WA*

- Shipped an **org-wide plug-in for Microsoft's internal Copilot**, reducing engineers' triage times by **60%**.
- Analyzed **1,000,000+ Word crashes daily** with Kusto and OpenTelemetry by building a reliability AI agent.
- Fixed Word bugs surfaced by agent's output, **eliminated 50,000+ enterprise-facing crashes over 2 releases**.

Synopsys January 2026 – April 2026

Software Engineer Intern *Pittsburgh, PA*

- Designed and implemented Ansys Mechanical's RAG coding agent memory solution with semantic clustering.
- Built chatbot's full test suite and a multi-turn eval harness with bucket visualization on Ansys test infrastructure.
- **Raised multi-turn accuracy from 40% to 82%** by shipping Go code to production behind feature flags.

PNC Financial Services May 2025 – August 2025

Software Engineer Intern *Pittsburgh, PA*

- Built a full-stack **housing analytics app** that turned **1M+ housing records** into homeowner-facing insights.
- Authored Angular unit tests, raising coverage from 45% to 85% and cutting production bugs by 35%

University of Pittsburgh August 2024 – April 2025

Teaching Assistant *Pittsburgh, PA*

- Led recitations and practices for 100+ students in Discrete Math, **raised average final scores from 78 to 84.**
- Held 5+ hours of weekly office hours, coaching students on **proof techniques and algorithmic thinking**.

Projects

DannyTorch | *Python, NumPy*

- Designed deep learning framework in pure NumPy **without AI**, building an autograd engine over a dynamic graph
- Implemented full transformer stack (MHA with causal masking, RoPE, RMSNorm, SwiGLU) behind a module API
- Verified all autograd ops against gradients, then trained a small LM on one A100, **cutting loss from 4.9 to 1.2**

GhostPIN | *Node.js, Redis, PostgreSQL, WebAuthn* **2nd Place, HackHarvard 2025**

- Designed an anti-phishing SDK with RSA-signed protocol and origin-bound nonces that block fake checkout pages.
- Integrated biometric MFA into checkout flow via WebAuthn, with approvals verified end-to-end in under 200ms.

Mantis | *Python, OpenCV, YOLOv8, GCP, Twilio* **2nd place, CalHacks 2025**

- Built a full-stack surveillance platform detecting theft and weapons in live feeds with real-time owner alerts.
- Architected a Python detection core streaming over WebSockets into a reactive Next.js/Convex dashboard on GCP.

Technical Skills

Languages: Python, C/C++, Java, Go, TypeScript, JavaScript, SQL, OCaml
Machine Learning & AI: PyTorch, NumPy, HuggingFace, transformers, model evals, benchmarking
Cloud & DevOps: AWS, GCP, Azure DevOps, Docker, Linux, Git
Backend & Frontend: Spring Boot, Kafka, Redis, React, Angular

Leadership

Director January 2025 – September 2025

SteelHacks *Pittsburgh, PA*

- Grew the event **21%** year over year to 600+ attendees and raised $50K+ from Google, Amazon, and PNC.

Technical Team Member September 2026 – Present

Google Developer Group, Budapest University of Technology and Economics *Budapest, Hungary*

- Led team of engineers to build myBME for students and co-organize guest tech lectures **while studying abroad.**